

Critical paper review

How to read a paper and write a critique

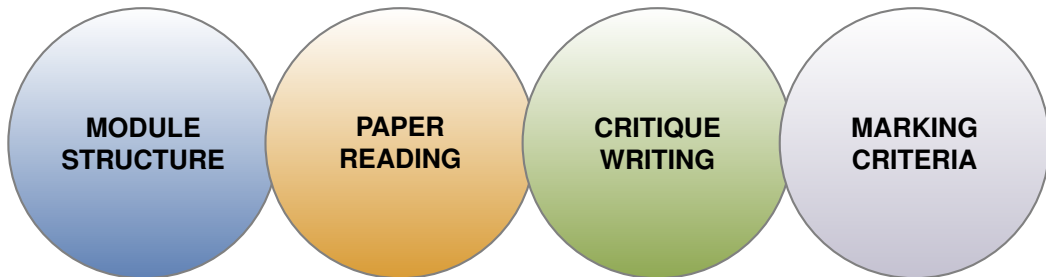
Dr. Ken Pierce
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School of Computing Science



CSC2034 — Contemporary Topics in Computing
2026-02-09

Outline



PART I

Module structure

General lectures

- 1 Module overview
- 2 Assignment 1: reflective log
- 3 **Assignment 2: article review**
- 4 Finding and reading academic sources [L]
- 5 Assignment 3: project
- 6 Finding information for your project [L]
- 7 Technical writing [L]
- 8 Research chats (TBC)

Easter break

- 9 Research chats (TBC)
- 10 Typesetting reports with $\text{\LaTeX}$

FDC G.41

every **Monday**
@14:30

Coursework

- **Reflective log** (20%) — due immediately after taster weeks
 - one entry per topic (≈ 256 words)
 - reflection on the experience and lessons learned
- **Article review** (20%) — due in the middle of March
 - structured critical review (max. 1k words)
 - three articles per topic to choose from
 - project independent
- **Project** (60%) — due in the first week of May
 - mini-dissertation / report format (max. 2.5k words)
 - specific task in one of the topics, assigned based on preferences

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PART II

Paper reading

What is in a paper?

A scientific paper is a formal document that presents original research findings and contributes to the body of knowledge in a specific field of study.

- 1 Abstract
- 2 Introduction
- 3 Methods
- 4 Results
- 5 Discussion
- 6 Summary

Headers and order will vary depending on domain/venue

Visualizing the Hidden Activity of Artificial Neural Networks

Paolo E. Roudier, Samuel G. Fiedel, Alexandre X. Falcão, and Alessandro C. Silva

Abstract.—In machine learning, pattern classification assigns high-dimensional vectors (information) to classes based on generalization from examples. Artificial neural networks currently achieve state-of-the-art results in this task. Although such networks are typically used as black boxes, they are also widely believed to learn (high-dimensional) higher-level representations of the original observations. In this paper, we propose using dimensionality reduction for two tasks: visualizing the similarities between learned representations of observations, and visualizing the relationships between artificial neurons. Through experiments conducted in three benchmark image classification benchmark datasets, we show how visualization can provide highly valuable feedback for network designers. For instance, our discussion in one of these datasets (SVHN) includes the presence of interpretable clusters of learned representations, and the partitioning of artificial neurons into groups with apparently related discriminative roles.

Index Terms.—Artificial neural networks, dimensionality reduction, algorithm understanding

1 INTRODUCTION

In machine learning, advances in computing power and techniques for building and training (deep) artificial neural networks (ANNs) have allowed these models to achieve state-of-the-art results in many applications related to pattern recognition [39]. However, successfully training ANNs is a primarily time-consuming, and requires significant expertise [31].

In data visualization, dimensionality reduction has been successfully used to compute projection representations of high-dimensional data in lower-dimensional (usually 2D) spaces that try to preserve the data structure. This structure is related to the data distribution, relationships between points, and presence of clusters [28, 24].

When depicted by scatterplots, projections can be used to reason about the original data. Compared to other high-dimensional data visualizations, such as scatterplots, parallel coordinates, star coordinates, and their variants, projections are considerably more scalable with respect to the number of dimensions [23, 44]. They are also scalable with respect to the number of observations, although visual clutter can eventually become problematic. Consequently, computational and recent dimensionality reduction techniques are able to deal with hundreds of thousands of observations at interactive rates [33, 19].

In this paper, we demonstrate the potential of dimensionality reduction techniques to provide insightful visual feedback about ANNs. Although we focus on multilayer perceptrons and convolutional neural networks, our approach is extensible to other types of networks (e.g., LSTM or Recurrent networks [14]). Specifically, our proposed visualizations address the following two tasks:

- T1: Exploring the relationships between alternative representations of observations learned by ANNs.
- T2: Exploring the relationships between artificial neurons.

The projection-based visualization approach that we propose for T1 is previously found in the machine learning literature [7, 29, 41, 36, 30].

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However, such projections are typically used for illustrative purposes (Sec. 3). In contrast, we show how projections can aid existing approaches for understanding and improving ANNs (Sec. 3). Specifically, using three widely studied benchmark image classification datasets, we show how our visualization approach is able to confirm facts that are already known about ANNs, and reveal previously unseen relationships between learned representations. In this context, we also propose a novel visualization of the evolution of such representations (Sec. 3.6).

Our approach towards T2 is completely new in the context of ANNs, although it is related to techniques developed for feature space exploration (Sec. 3). Similarly to our approach for T1, we use projections to represent similarities between artificial neurons (given a particular set of input observations). We also propose a novel visualization of the relationships between artificial neurons and classes (Sec. 4.2). Although presented separately, our visualization approaches for T1 and T2 should be seen as complementary for understanding ANNs, as exemplified in Sec. 4.5.

This paper is organized as follows. Section 2 briefly introduces pattern classification, together with the notation and definitions used in the text. Section 3 relates this paper to previous work in machine learning, information visualization, and visual analytics. Section 4 details the protocol followed by our experiments, which, although not crucial to our claims, allows for reproducibility. Section 5 presents the results of our projection-based visualizations of the relationships between learned representations for different datasets (T1), highlighting valuable insights gained from visualization. Section 6 presents our projection-based visualizations of relationships between artificial neurons (T2). Section 7 discusses the limitations of our work. Section 8 summarizes our findings and suggests future work.

2 PRELIMINARIES

A dataset $\mathcal{D}$ is composed of pairs (x, y) , where $x \in \mathbb{R}^n$ is an observation, and $y \in \{0, 1\}^c$ is a target class assignment. If $y_0 = 1$, observation x belongs to class c . In this paper, each observation corresponds to a 2D image and belongs to a single class, although there are no limitations of our proposal.

We consider two types of ANNs: multilayer perceptrons (MLPs) and convolutional neural networks (CNNs). Such a network represents a parametric function $f: \mathbb{R}^n \rightarrow \{0, 1\}^c$, which usually attempts to generalize class assignments from examples in $\mathcal{D}$. Computations in these networks is performed by artificial neurons, which are typically organized into layers, as outlined next.

Multilayer perceptrons. In this model, the weighted input to neuron j in layer l is calculated as $\sum_{i=1}^n w_{ij}^{(l)} x_i^{(l-1)}$, where $x_i^{(l-1)} \in \{0, 1\}$ is an input parameter, and $w_{ij}^{(l)} \in \{0, 1\}$ is the activation (output) of neuron i in layer $l-1$ (0FF). In other words, each neuron computes a linear combination, plus a bias, of neuron activations from the previous

Reading a paper

- You don't have to read a paper in order!
- Read the Abstract — sometimes that's all you need
- Take a break!
- Read the Introduction — what do the authors say they are going to do?
- Read the Conclusions — what do the authors think they've shown?
- Take another break...
- Read the main body — does it support the conclusions?

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Tips for academic reading

DOs

- Read effectively, efficiently, and strategically
- **Skim** and **scan** texts before reading in depth
- Exercise **critical thinking** when researching
- Focus on authentic, authoritative, and **academic information**

DON'Ts

- Try to read everything in order like a novel
- Trust everything you read without thought
- Rely on large language models — **they hallucinate**

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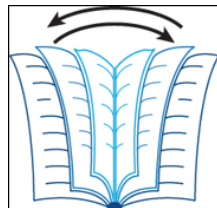
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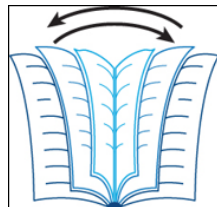
Skimming

- A **superficial, fast** activity — **flicking through a book**
- Acquire a 'big picture'
- It's **not cheating**
- Looking for **relevant** information **quickly**
- What to skim?
 - Titles, subtitles, contents pages, indexes, headings, ...
 - Figures, diagrams, plots
 - Abstracts, introductions, and conclusions
 - Topic sentences
- **Be fast. Scroll. Annotate.**



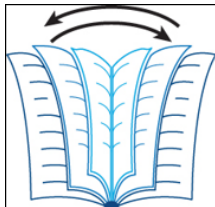
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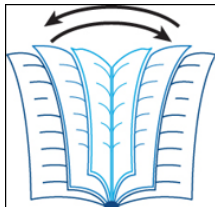
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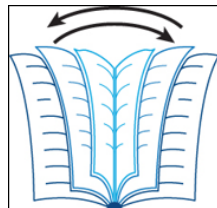
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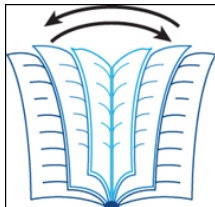
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- 2 Identify key **sites of meaning** in sentences
- 3 Look for **topic**, **action**, and **detail**:

Despite its popularity, **social media** has become an insidious, negative and highly addictive means of **disseminating** 'fake news.'

- 4 Elements
 - Topic: social media
 - Action: disseminating
 - Detail: fake news



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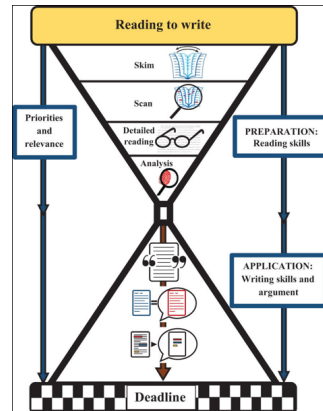
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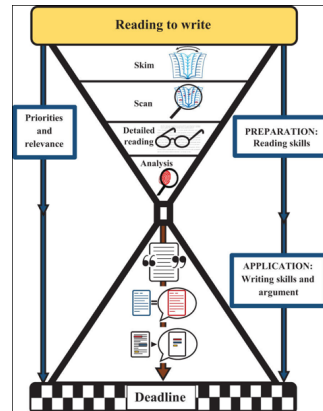
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- 2 Both important skills to practice
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- 6 Return to detail once you have 'big picture'



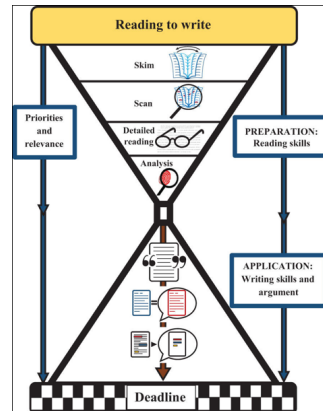
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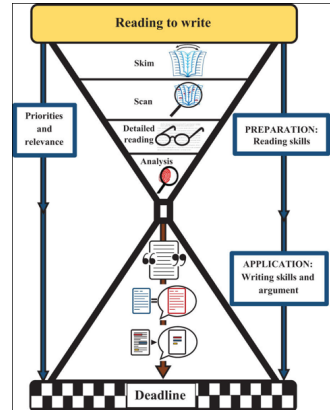
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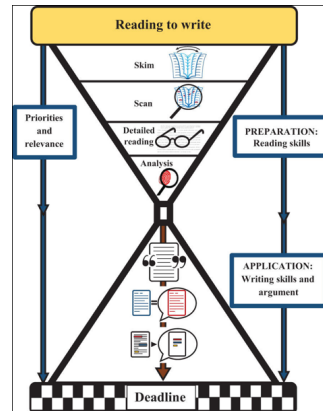
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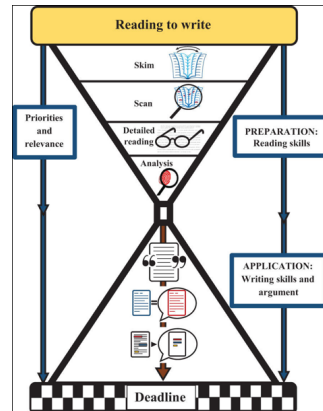
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PART III

Critique writing

Writing the review

- Canvas quiz guides the review
- Nine questions covering the following:
 - 1 Which paper you picked
 - 2 Introduction
 - 3 Objectives
 - 4 Methods
 - 5 Results
 - 6 Discussion
 - 7 Research context
 - 8 Summary
 - 9 Reference

criticise
verb

- indicate the faults of something in a disapproving way.
- **form and express a judgement of a scientific or artistic work.**

Thinking about research context

The 'W' questions

- **Who** conducted the research? What were/are their interests?
- **When** and **where** was the research conducted?
- **Why** did they do this research?
- Does the research have a broad (even global) relevance?

The context

- Were others pursuing **related research** at the time the work was done?
- Who funded the research? Did the **funder** influence the research?
- On what **prior observations** was the research based?
- How **important** was the research question posed by the researcher?

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Thinking about research significance

The effect of the research

- How did other researchers view the **significance** of the research?
- Did the research lead to **new questions** or hypotheses?
- Have other researchers **supported** or refuted the research?
- Did the research make a significant **contribution** to human knowledge?
- Did the research produce any **practical** applications?
- What are the social, political, technological implications?

Look at articles that cite this work

PART IV

Marking criteria

Requirements

You should:

- select **one** of the articles and **read** it carefully
- follow the guidance to help with **analysis**
- answer the questions to complete your review

Marking criteria:

- Content and style — relevant, accurate, error free
- Summarising — present idea, comment on organisation and presentation
- Analysis — complexity, perceptiveness, originality, and depth of thought
- Use of evidence — focus, coherence, insightful,

Summary

Critical review assessment:

- gives you experience **reading** about **technical subjects**
- lets you **engage** with one of the topics further
- help you to practice **critically evaluating** and source
- gives you experience **writing** a preview of a **scientific paper**

Next

New topic intro lecture: **Human Computer Interaction**

Tommorrow

1h lecture (FDC @ **15:30**) + 2h practical session on **design requirements**

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